

Estimating the heterogeneous effects of a utility-scale solar facility on house prices[☆]

Michael Cary^{a,*,}, Susan Chen^b, Jessica Douthit^c, Njeri Jackson^d, Michael McCloud^e, Benjamin Rosner^{f,g}

^a Department of Agricultural and Applied Economics, Virginia Tech, United States of America

^b Department of Agricultural Sciences, Clemson University, United States of America

^c Department of Computer and Information Science, Berea College, United States of America

^d Department of Human Development and Family Sciences, Virginia Tech, United States of America

^e Department of Statistics, Virginia Tech, United States of America

^f Department of Economics, Virginia Tech, United States of America

^g Department of Geography, Virginia Tech, United States of America

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ABSTRACT

Utility-scale solar facilities are being installed across the United States at a growing rate. However, the effects of these facilities on local economies are not well studied, especially in rural settings where these facilities are often located. In this paper we estimate the causal effect of a new utility-scale solar facility opening on nearby house prices using a distance-decay difference-in-differences house price hedonic model in the context of Prince George County, Virginia. We find that more expensive houses decreased in value by at least 2%, while less expensive houses did not. Explaining this effect, we also find evidence of a decreased transaction volume in neighborhoods near the newly constructed utility-scale solar facility, likely due to decreased demand for these houses. These findings offer profound implications for the optimal siting of utility-scale solar facilities in the future from a socio-economic perspective.

1. Introduction

Reliance on renewable energy has increased at a remarkable rate, with a 50% global surge in renewable energy capacity additions between 2022 and 2023 (International Energy Agency, 2024). Due to their comparably low cost, utility-scale solar and wind-powered facilities have been the most popular, and are subsequently anticipated to account for 96% of renewable energy capacity additions in the next five years (International Energy Agency, 2024). Due to both the U.S. Inflation Reduction Act and individual state efforts, the United States has experienced the third-largest growth-rate globally (International Energy Agency, 2024). One such state effort is the Virginia Clean Economy Act, which requires all state energy production to be renewable by 2050 (Virginia Department of Energy, 2024). As of 2022, 11% of Virginia's electricity came from renewable sources. Solar power was the most prominent form of renewable energy and accounted for 6% of all state energy production in 2022. Between 2020 and 2022 solar energy production doubled (Energy Information Administration, 2024).

Despite global, national, and state compliance with renewable energy initiatives, local resistance to utility-scale solar facilities is common (Scovell et al., 2024). Utility-scale solar facilities (USSFs) are large-scale commercial facilities that supply electricity to the commercial grid. When deciding the optimal location of a USSF, factors beyond proximity to the grid are critical. These factors include parcel size, soil characteristics, topography, proximity to roadways large enough to allow for heavy machinery to access the parcel, and how the parcel is zoned (Bobrowski and Łaska, 2023; Zambrano-Asanza et al., 2021). They are usually located on large tracts of land at the edge of suburban areas. As such, they affect two constituents: residential homeowners in suburban locations and farms on the rural outskirts. For residential homes in suburban locations, the main objections to USSFs are the potential negative effects on residential property values due to the visual deterioration and environmental degradation of the surrounding landscapes. In rural areas, additional concerns range from decreasing the supply of (farm)land and the potential upward pressure on already high (farm)land prices to maintaining the region's rural culture.

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* Corresponding author.

E-mail address: macary@vt.edu (M. Cary).

To address this concern, some communities have dedicated local ordinances to restricting USSF developments. The housing market studied in this paper, that of Prince George County, Virginia, recently put into place an upper bound of 2.74% of its total land area for how much land can be allocated to USSF developments.¹ This amounts to approximately 4603.5 acres (1863 hectares) of land that can be allocated to solar energy generation. However, this does not affect our analysis as no houses in our sample were within six miles of any other USSF development in Virginia, Prince George County or otherwise. In fact, the grand total of all projects proposed to date does not meet this 2.74% limit.

These myriad concerns suggest a heterogeneous effect on residential prices in suburbia versus rural areas. For suburbia, the visual/environmental disamenity may have a negative effect on property values. At the same time, for more rural areas, there may be an increase in the price of (farm)land as a result of a decrease in land supply, which drives property prices upwards. Additionally, once the potential for USSFs has been established in the region, there may be an increase in demand for land for further USSF development. To date, very few studies have examined the effect of USSFs on housing prices. [Elmallah et al. \(2023\)](#) used data from six states across the US and found a negative housing price effects, while [Hao and Michaud \(2024\)](#) found a positive effect on home values, also using a multi-state sample.

Given the recency and incongruities of this literature, there is not yet an academic consensus on the effects of USSFs on house price. However, the economic theory of spatial equilibrium suggests one significant reason these opposing results have been found. Under spatial equilibrium, amenities, especially natural amenities, are capitalized differently across different housing markets ([Roback, 1982](#)). By combining several housing markets into one hedonic model of house prices, the estimated effects of USSFs on house prices are likely to be biased. While a high-dimensional fixed effect model, such as that used by [Elmallah et al. \(2023\)](#), can help to mitigate the consequences of combining multiple housing markets into one hedonic model, combining multiple housing markets in this way can still bias an estimate of a nationwide mean effect on house prices from newly constructed USSFs. For this reason, our paper estimates the effect of a newly constructed USSF on house prices in the context of a single housing market where (dis)amenities – like USSFs – are not subject to heterogeneous capitalization, thereby eliminating this potential source of bias.

In addition to studying the overall effect of a USSF on house prices, we want to better understand the underlying mechanisms driving these effects. To do this, we perform two additional analyses. First, to determine if there are heterogeneous price effects for different types of houses, we split our sample into two subsamples — those houses sold at or above the median price of our sample and those houses sold below the median price. This allows for an improved understanding of the heterogeneous effects of USSFs on house prices that [Elmallah et al. \(2023\)](#) found. Second, to determine if USSFs affect the demand for housing near the USSFs, we study the effect of a USSF on housing turnover at the census block-year level. This makes our paper the first in the literature to study the impact of new USSF construction on housing turnover in any setting.

Our paper estimates the impact of the newly constructed Caden Energix Rives Road Solar Facility on house prices in a single housing market — that of Prince George County, Virginia. This USSF, the Caden Energix Rives Road Solar Facility, was the first USSF development project in Prince George County. Opening in May 2020, this facility covers 147.6 acres (59.73 hectares) and has a power generation capacity of 19.7 MW.² As explained earlier, the net effect on housing

prices is the sum of the disamenity effects and the increase in value of land. Since these represent opposing price effects, the net effect is theoretically unclear. By measuring the impact of a USSF on the price of nearby residential properties, we examine whether a USSF is on net an amenity or a disamenity. By doing so in a single housing market, our estimate is free from the bias generated by combining multiple markets into one hedonic model. Furthermore, our setting is a community on the exurbs of a metro area (Richmond, Virginia), and is both suburban and rural in its composition.

To estimate the causal effect of a USSF on house prices, we use a distance-decay difference-in-differences house price hedonic model. This model accounts for both the effect of a USSF on house prices and whether the effect varies by distance from the facility. Our results show that houses closer to the USSF experienced an increase in price by approximately 6%. We also find that for a sample of more expensive homes, the USSF decreased property values, which attenuates by distance from 9% to 2.3%. Our housing turnover analysis supports these attenuated results, showing that the demand for houses closer to the USSF decreased after its construction relative to the control homes. Taken together, these results suggest that the opposing forces affecting house prices, both the disamenity effect that diminishes house prices and the increased demand for land that can drive up property values, may have more prominence with homes at the higher end of the market.

The remainder of this paper is structured as follows. First, we will review the relevant literature on the consequences of constructing USSFs, emphasizing house prices. Then, we will present our data and methodological approach in detail. After that, we will present and discuss our results before concluding this paper with a detailed discussion of the policy implications of our findings.

2. Literature review

USSFs are becoming increasingly more common in the US, mainly because solar energy represents a renewable energy source that drastically reduces the CO₂ emissions from energy generation ([International Energy Agency, 2024](#)). Despite these benefits, there is a tradeoff at play. USSFs are viewed as having disamenities and opposition to USSFs is strong in rural communities, which are typically best suited for USSF development ([Roddis et al., 2020](#)). These disamenities range from environmental concerns such as increased runoff in certain situations ([Cook and McCuen, 2013](#)), to the facility itself being a visual disamenity ([Otero et al., 2022](#); [van den Berg and Tempels, 2022](#)), to providing upward pressure on land prices. Importantly, concern over these disamenities is directly related to proximity to USSFs ([Carlisle et al., 2016](#)).

To best address these and similar concerns around USSF development while promoting solar energy development, state governments in the US have gone as far as implementing protectionist policies designed to keep out-of-state facilities from participating in their Solar Renewable Energy Certificates (SREC) markets ([Cohen et al., 2022](#)) while at the same time, providing tax breaks to USSF developers within their states ([Cohen et al., 2020](#)) and to individual households to install solar panels ([Best et al., 2019](#)). With an arguable exception of encouraging household uptake of rooftop solar panels, an additional means to generating clean energy ([San-Martín and Elizalde, 2024](#)), these policies do not address the disamenities of USSFs.

The very brief literature on the effects of these facilities on nearby house prices has found both positive and negative house price effects associated with USSFs. For instance, [Hao and Michaud \(2024\)](#) found that average house prices in counties increased by as much as 2% after USSFs were constructed. However, this paper relied on average home price estimates from Zillow rather than actual data on house transactions. Conversely, houses within one km of a solar facility in the Netherlands decreased on average by 2.6% after the facility opened ([Dröes and Koster, 2021](#)). A study in Britain found an even

¹ https://www.princegeorgecountyva.gov/planning_and_zoning/solar_energy_facilities.php

² <https://eerscmap.usgs.gov/uspvdb/viewer/>

larger 5.4% decrease in house prices (Maddison et al., 2023). For reference, these effects are similar in direction to the impact of opening new fracking wells (Stephens and Weinstein, 2019), and in contrast with the price effect of rooftop solar panels on house transactions, which has shown to be positive in the sun-intensive state of Arizona (Qiu et al., 2017), although renters do not appear willing to pay more to live in a house equipped with rooftop solar panels (Hammerle et al., 2023).

These negative price effects on houses can be very heterogeneous. For example, Gaur and Lang (2023) found greater negative effects in rural areas than in urban areas for the neighboring US states of Massachusetts and Rhode Island, which were significant in both cases. This urban–rural distinction is rather interesting because land values have been shown to increase after USSF development is introduced to the regions, assuming the land is sufficiently close to transmission lines so that it has the economic potential for USSF development (Abashidze and Taylor, 2023). Moreover, land transitioning to USSF development decreases the supply of farmland available. It also decreases the suitability of the land for future agricultural purposes (Abashidze and Taylor, 2023). A negative net effect for rural properties suggests that rural communities view the combined visual disamenity and the diminished agricultural productivity of the land as a net disamenity compared to urban communities. The negative effect in rural areas is likely due to concerns that solar facilities represent a threat to their agricultural livelihood and to their rural culture and heritage (Swanson et al., 2025). Similarly, Elmallah et al. (2023) found an average decrease of 1.5% in a multi-state study in the US, but in their case, this effect was driven primarily by a decrease in rural property values.

Our paper builds on this literature in two ways. First, by studying a single housing market that is both suburban and rural, we can provide more nuanced insight into this heterogeneity while also avoiding the issue of a potentially biased estimate due to heterogeneous capitalization of amenities under spatial equilibrium. Second, we are the first paper to directly estimate how housing turnover changes in response to the opening of a USSF. This allows us to discuss the underlying economic mechanisms explaining the decrease in house prices after a USSF opens.

3. Data

Our house transactions data comes from Vision Government Solutions and includes data on all residential property transactions in Prince George County, Virginia, from the beginning of 2012 through the end of Q1 2024. This data contains transaction prices and dates, the addresses of properties, the names of the involved parties, a standard set of housing characteristics including square footage, acreage, the number of rooms, bedrooms, bathrooms, etc., and the specific neighborhood each house belongs to. Summary statistics of the final data set, which consists of 1640 observations, can be found in Table 2.

To determine whether or not a house will be considered as treated in this study, we used geospatial data from the U.S. Large-Scale Solar Photovoltaic Database published by the US Geological Service. By geolocating each house using its address, we computed the distance from each house to the USSF in miles. We then created a series of treatment variables based on how close houses are to the solar facility. Specifically, we created indicator variables for houses between zero and one mile, between one and two miles, and between two and three miles from the solar facility. While our preferred model specification will be a distance-decay difference-in-differences house price hedonic model, we also run a standard difference-in-differences model, thus we created a standard treatment variable encompassing all of the above radii.

Similarly, we use geospatial data from the US Census Bureau (TIGER shapefiles) on the location of bodies of water — both lakes and rivers. These are key amenities affecting house prices in Prince George County. Bodies of water and waterfront properties are common in Prince George County, with the James River being the most notable in the county. Since bodies of water are commonly used for recreational activities

(fishing, boating, etc.) and are also an important visual amenity, proximity and access to water increases the value of properties. Thus we control for the distance to the nearest lake and river for all parcels in our sample.

In addition to the creation of these variables, we adjust all sales prices to real 2024 US dollars using the CPI. With the creation of new variables completed, we also performed standard data cleaning for hedonic modeling. This included removing the top and bottom percentile of all transactions based on the real value of the transaction and removing arms-length transactions from the sample. We removed historical properties, defined as houses over 150 years old. Such properties are common in Prince George County due to its close proximity to Jamestown, the location where the British first began colonizing the American continent. Table 1 shows summary statistics for the entire data set, and Table 2 compares the treated and control units; t-tests confirmed that there are no statistically significant differences between the treated and control houses in our sample.

4. Methodology

4.1. Baseline distance-decay difference-in-difference model

To estimate the effect of constructing the Caden Energix Rives Road Solar Facility on nearby house prices, we use a difference-in-difference house price hedonic model. The Caden Energix facility was chosen because it allows us to study a single, well-defined housing market in which the “law of one price” is satisfied, i.e., that identical houses will sell for the same price, which does not occur when multiple, distinct housing markets are amalgamated (Waights, 2018). Our preferred specification is a distance-decay difference-in-difference house price hedonic model, but for thoroughness, we compare this model against a standard difference-in-difference house price hedonic model.

We use a set of three difference-in-difference estimators to estimate the effect of facility on housing prices. The distance variable is parsed into three bins — the first is defined by being within one mile of the facility, the second by being between one and two miles from the facility, and the third is defined by being between two and three miles from the facility. This follows the increasingly common methodological approach used in papers studying the impacts of USSFs such as Hu et al. (2025). More generally, distance-attenuating effects have long been estimated in environmental economics (Kiel and McClain, 1995; Palmquist et al., 1997). The USSF along with the houses in our sample is presented in Fig. 1 for the time period under study. Houses within one mile of the facility are colored red, houses between one and two miles are colored orange, houses between two and three miles are colored yellow, and houses between three and six miles (our control group) are colored black.

The post variable is a binary variable set to one for sales from May 2020 onward and zero otherwise. In the formal specification in Eq. (1) below, $D_{i,t}^j$ denotes the interaction between treatment and post. We are interested in estimating the effect of constructing the Caden Energix Rives Road Solar Facility on the prices of houses near the facility relative to those further away. This effect is represented in our model by the vector θ_j , which estimates the effect of proximity of the solar siting facility on housing prices. Specifically, the vector θ estimates the effect within one mile, from one to two miles, and two to three miles, respectively.

In addition to the control variables presented in the previous section, we also include neighborhood (η_n) and year (δ_t) fixed effects to account for variation in neighborhood quality and time trends in this

Table 1
Summary statistics.

Variable	Mean	Std. Dev.	Minimum	Maximum
Price	251,922.480	97,376.549	78,649.357	752,181.084
Treated	0.466	0.499	0.000	1.000
Post	0.337	0.473	0.000	1.000
Age (Years)	33.624	13.676	0.000	96.000
Square Feet	1,818.107	617.193	492.000	5,767.000
Rooms	6.570	1.507	3.000	14.000
Bedrooms	3.305	0.577	1.000	8.000
Full Baths	1.898	0.588	1.000	5.000
Half Baths	0.502	0.522	0.000	2.000
Fireplaces	0.748	0.559	0.000	3.000
Acres	2.200	5.879	0.240	129.500
Distance to River (Miles)	5.427	2.234	0.316	11.282
Distance to Lake (Miles)	1.704	0.988	0.020	3.570

This table shows the summary statistics (mean, standard deviation, minimum, and maximum) for our data set. There are 1640 house transactions included.

Table 2
Comparison of treated and control groups.

Variable	Treated x Post	Control
Price	279,640.047 (89,156.562)	247,250.315 (97,954.347)
Age (Years)	38.949 (13.360)	32.727 (13.529)
Square Feet	1,748.772 (599.311)	1,829.794 (619.600)
Rooms	6.464 (1.569)	6.588 (1.496)
Bedrooms	3.371 (0.636)	3.294 (0.566)
Full Baths	1.814 (0.637)	1.912 (0.578)
Half Baths	0.536 (0.533)	0.496 (0.520)
Fireplaces	0.738 (0.544)	0.749 (0.561)
Acres	1.906 (4.175)	2.250 (6.119)
Distance to River (Miles)	5.679 (1.430)	5.384 (2.340)
Distance to Lake (Miles)	1.213 (0.717)	1.786 (1.003)
Observations	235	1405

Comparing group means between the treated and control units. Standard deviations are provided in parentheses. None of the variables exhibit a statistically significant difference between treated and control units.

housing market.³ The subscripts i , n , and t , denote individual properties, neighborhoods, and the year a property was sold, respectively.

$$\log(\text{Price}_{i,n,t}) = \sum_{j=1}^3 \theta_j D_{i,t}^j + X_{i,n,t} \beta + \eta_n + \delta_t + \epsilon_{i,n,t} \quad (1)$$

The primary concern with a difference-in-differences house price hedonic model is a violation of the parallel trends assumption, which in our case asserts that, had the Caden Energix Rives Road Solar Facility never been built, any difference in prices between treated and control houses would have remained constant post-treatment. In the context of our study, house prices near the facility before the facility was constructed were slightly lower than houses further away (see Table 9 in the appendix). To address this concern, we estimated an event study specification with the treatment group consisting of all houses within three miles of the facility (all houses from each of the three

treatment groups in the distance-decay model above). The results from this analysis are shown in Figure 5 in the appendix. As the graph shows, the parallel trends assumption holds for our study. Moreover, the two-way fixed effects specification in our difference-in-differences model can account for the fact that houses closer to the facility were prices slightly lower than houses further away. Thus, the difference-in-difference estimators in our model will yield unbiased estimates of the effect of constructing the Caden Energix Rives Road Solar Facility on house prices in Prince George County, Virginia.

4.2. Housing turnover

To better understand what drives the negative effects we find, we also examine housing turnover. This is an important addition to our main analysis because changes in turnover affect housing demand. If this solar facility is a (net) disamenity, then we should expect to see decreased turnover and demand for houses closer to the facility after the facility was constructed. This would not only establish that the solar facility is a net disamenity but would also provide an economic mechanism explaining the decrease in house prices.

³ We do not include neighborhood-by-year fixed effects because the neighborhoods (roughly 50) within six miles of the solar facility are homogeneous.

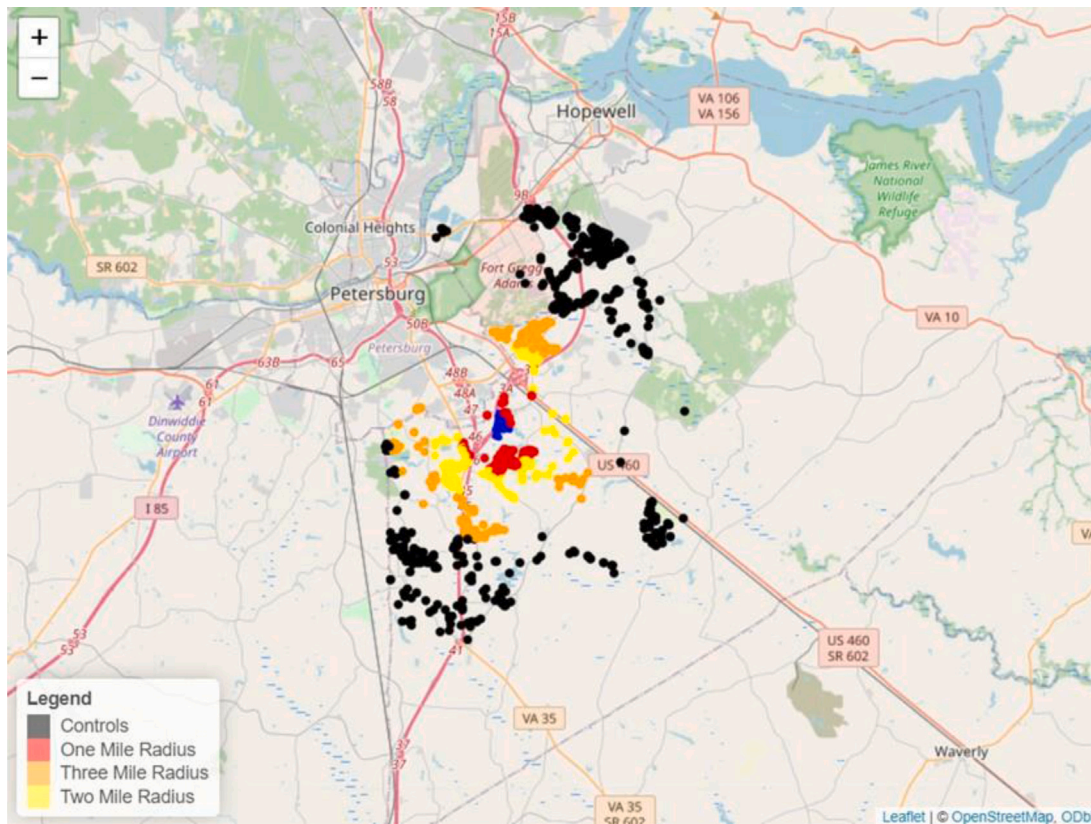


Fig. 1. Locations of House Transactions Relative to the USSF.

This figure shows the location of the Caden Energix Rives Road Solar Facility and the houses in our sample. The solar facility is colored blue. Treated houses are colored red, orange, or yellow depending on their distance from the facility. Control houses are colored black. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

We estimate the turnover model using a similar specification to (1), with the dependent variable now being the number of houses sold in each census block each year. Thus, this model uses the same universe of house transactions from the previous analyses, but with counts of transactions per census block per year as our dependent variable. The exact specification is shown in Eq. (2) where we now also include census tract-by-year fixed effects ($\chi_x \times \delta_t$) and census block level fixed effects (ζ_b) where Census blocks are considered treated based on the distance between the centroid of the census block and the solar facility. The subscripts b , x , and t , denote census blocks, census tracts, and years, respectively. In total, 58 of the 160 census blocks within six miles of the solar facility are considered treated. Pre-trend testing results support the parallel trends assumption and are shown in Figure 6 in the Appendix.

$$Sales_{b,x,t} = \sum_{j=1}^3 \theta_j D_{j,t} + (\chi_x \times \delta_t) + \zeta_b + \epsilon_{b,x,t}, \quad (2)$$

5. Results and discussion

5.1. Results from the baseline specification

In Table 3, we present the results for the empirical model from Eq. (1). The first three columns use a standard difference-in-differences approach, with treatment defined as being within one, two, or three miles of the facility, respectively. In each of these models, the control group consists of houses in the region from the edge of the treated region to six miles out from the solar facility. The fourth column in each schedule of regressions is our preferred distance-decay difference-in-differences model with separate treatment effects for houses between zero and one mile from the solar facility, one and two miles from the solar facility,

and two to three miles from the solar facility. While not displayed here, all models control for the full set of control variables described in Table 2, as well as a quadratic specification for age included to address the nonlinear relationship between age and price. These results suggest that we can attribute an approximately 6% increase in house prices after the installation of the solar facility for houses within two miles of the facility.

We also examine whether the effects differ between the upper and low end of the price distribution. We separate the sample into higher-priced (at or above the median home price) and lower-priced homes (below the median). We present these results in Table 4. For lower-priced homes, the effect of the solar facility is positive but statistically insignificant. For higher-priced homes, we find statistically significant and negative effects. For the standard difference-in-differences estimator with a two-mile treatment radius, we see a statistically significant decrease of over 6%. For the distance-decay model, we find statistically significant and negative effects that do indeed decay as we move from smaller to larger treatment radii. For more expensive houses within one mile of the solar facility, we find roughly a 9% decrease in value relative to houses between three and six miles from the facility. For more expensive houses between one and two miles from the facility, we find an almost 7% decrease in value relative to houses between three and six miles from the facility. And for more expensive houses between two and three miles from the facility, we find a decrease in value of 2.3% relative to houses between three and six miles from the facility.

Our results for more expensive houses align with previous estimates in the literature, such as Dröes and Koster (2021) and Maddison et al. (2023). Our results contrast with those of Elmallah et al. (2023) insofar

Table 3
Distance-decay model regression results.

	<i>log(Price)</i>			
	1 mile	2 miles	3 miles	Decay model
Treated $\times$ Post	0.053*** (0.019)	0.064*** (0.019)	0.025 (0.017)	
Treated ($0 \leq D < 1$) $\times$ Post				0.059*** (0.018)
Treated ($1 \leq D < 2$) $\times$ Post				0.059** (0.030)
Treated ($2 \leq D < 3$) $\times$ Post				−0.016 (0.019)
Observations	1,640	1,640	1,640	1,640
# Treated Units	175	238	352	765
Adjusted R^2	0.652	0.654	0.652	0.653

This table contains the results using the model specified in Eq. (1) and the full sample of observations. All models here include the full set of control variables. Cluster-robust standard errors are provided in parentheses, with clustering occurring at the neighborhood level. Statistical significance is denoted by * for $p < 0.10$, ** for $p < 0.05$, and *** for $p < 0.01$.

Table 4
Heterogeneity analysis regression results.

	<i>log(Price)</i>							
	Lower Priced Homes				Higher Priced Homes			
	1 Mile	2 Miles	3 Miles	Decay Model	1 Mile	2 Miles	3 Miles	Decay Model
Treated $\times$ Post	0.030 (0.040)	0.050 (0.036)	0.027 (0.048)		−0.063 (0.042)	−0.063* (0.033)	−0.039** (0.018)	
Treated ($0 \leq D < 1$) $\times$ Post				0.037 (0.038)				−0.086* (0.044)
Treated ($1 \leq D < 2$) $\times$ Post				0.053 (0.059)				−0.066* (0.034)
Treated ($2 \leq D < 3$) $\times$ Post				−0.016 (0.095)				−0.023** (0.010)
Observations	818	818	818	818	822	822	822	822
# Treated Units	134	153	159	446	41	85	193	319
Adjusted R^2	0.170	0.170	0.170	0.167	0.724	0.725	0.724	0.725

This table contains the results from the heterogeneity analysis using the model specified in Eq. (1), with the sample split at the median real transaction price. All models here include the full set of control variables. Cluster-robust standard errors are provided in parentheses with clustering occurring at the neighborhood level. Statistical significance is denoted by * for $p < 0.10$, ** for $p < 0.05$, and *** for $p < 0.01$.

as we find that suburban properties were more affected than rural properties.⁴

Collectively, these results confirm the hypothesis that USSFs are a source of both amenities and disamenities for local residents. In terms of disamenities, our results show a clear decrease in value for the more expensive houses in our sample. This supports our claim – and the claim of the literature more broadly – that USSFs are viewed as a disamenity by residents. However, the effect on the less expensive houses in our sample was positive and similar in magnitude to the results for the full sample, though statistically insignificant. Since we observe a significant and negative effect for the more expensive houses, this suggests that some positive amenity is offsetting any disamenities. The most likely explanation for this is an increase in the value of land. Building a USSF decreases the supply of land. This puts an upward pressure on land prices, increasing the value of nearby properties. Simply, the disamenities affect the value of the house, while the amenity benefit is capitalized in the value of land. For less valuable properties on similar acreage, the land will contribute more to the total value of the property. This is likely why we see a negative effect from the USSF on the more expensive houses and a positive but insignificant effect for the less expensive houses.

To establish the robustness of our results, we have performed three additional analyses. First, we ran a continuous treatment version of

our difference-in-differences model where we interact the log distance from the USSF with the Post indicator. The results for the full sample and for each of our two subsamples align with our main results and are provided in Table 10 in the appendix. Second, we split our sample into pre-treatment and post-treatment eras and estimated the effect of proximity to the facility using the underlying two-way fixed effects specification from our main model. These results are again consistent with our main results and can be found in Table 11 in the appendix.

Lastly, we created kernel density plots comparing the pre- and post-treatment house price gradients for the full sample (Fig. 2) and each of the two subsamples (Figs. 3 and 4). These figures are the key to understanding our results. In Fig. 2 we see the pre-USSF and post-USSF lines essentially mirroring one another, with the post-USSF curve slightly above the pre-USSF curve until the three mile mark. This indicates a potential increase in value for nearby houses after the USSF was constructed and supports our choice of threshold for treated status in our models.

However, when we disaggregate our sample into two groups at the median, we observe very different dynamics. For the more expensive half of our sample (Fig. 3), we observe a substantial decrease in value due to the USSF, which slowly dissipates and disappears entirely at the three-mile mark. This is a strong visual illustration of the effect described by our distance-decay difference-in-differences model. However, we see a different story for the less expensive half of the sample (Fig. 4). Here we see evidence of an increase in price that disappears just before the three mile mark, but we do not see a reliable fit in the control group (the area three to six miles from the USSF). This poor fit between the pre-USSF and post-USSF curves for the control group (houses more than three miles from the USSF) implies that the control

⁴ Furthermore, our results are robust to removing larger properties. We estimated the specifications of samples that had transactions for properties on at least ten acres and then again at five acres removed and found nearly identical results in both cases. The results from these robustness tests are omitted for concision.

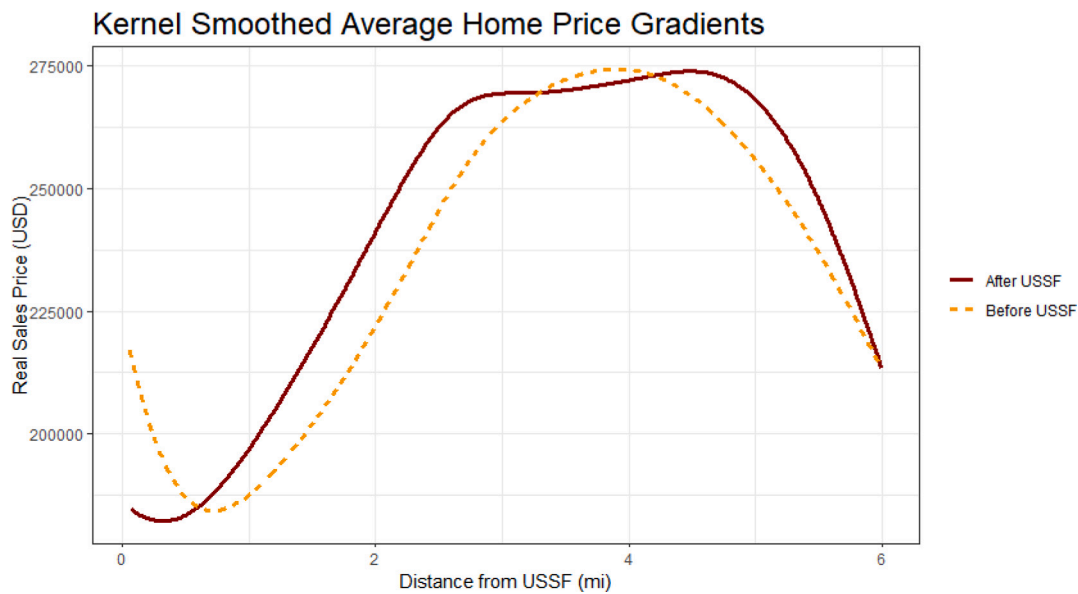


Fig. 2. This figure shows the price gradient with respect to distance from the USSF for the entire sample of house transactions for both the pre-USSF and post-USSF eras. The post-USSF curve is slightly above the pre-USSF curve until the three mile mark.

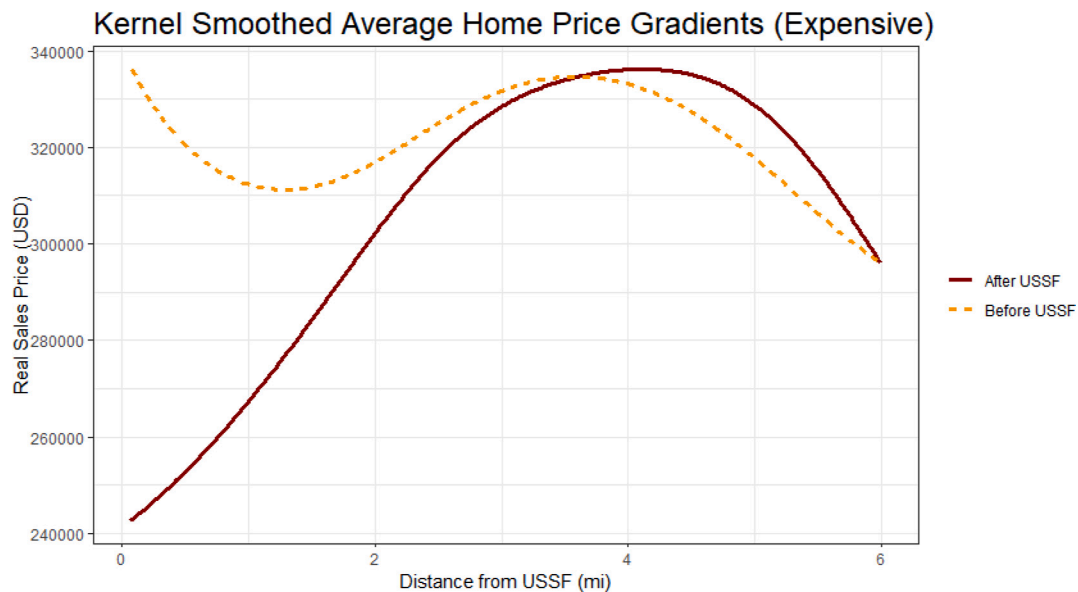


Fig. 3. This figure shows the price gradient with respect to distance from the USSF for the subsample of the more expensive (above the median) house transactions for both the pre-USSF and post-USSF eras. A marked decrease in price is observed after the USSF relative to before the USSF. This price decrease disappears almost exactly at three miles. After three miles the pre-USSF and post-USSF price gradients are nearly identical, suggesting that the effect of the USSF did not exceed the three mile radius used to define treated status in our models.

units do not constitute a reliable counterfactual for the treated units for this subsample. An inadequately comparable control group undermines the assumptions of causal inference. This casts doubt on the credibility of the results from the subsample analysis for the less expensive houses, and by extension, for the full sample analysis. Thus, when drawing conclusions from the results in this paper, we primarily rely on the subsample comprising the more expensive half of house transactions. We note that the price range for the more expensive subsample of house transactions are better aligned with national house prices.

Our results for the more expensive subsample also align closely with the previous literature. All studies discussed earlier that use a house price hedonic model to estimate the effect of USSFs on house prices find a significant and negative price effect. This is the case for both case studies of one single market and for those aggregated across

many housing markets. In all cases, the magnitude of the effect is a single digit percentage decrease in house prices. Per [Table 4](#), our results fall within this range. For example our estimated effect size comparing houses within three miles of the USSF to houses between three and six miles from the USSF for the higher priced houses in our sample is -3.9% (column seven of [Table 4](#)). This lies between what was found by [Dröes and Koster \(2021\)](#) and [Maddison et al. \(2023\)](#) and is similar to what was found by [Hu et al. \(2025\)](#). The only results in our literature review with positive estimated effects either estimated the effect on land in the absence of houses ([Abashidze and Taylor, 2023](#)), the effect of rooftop solar panels ([Qiu et al., 2017](#)), or used a house price index from Zillow rather than actual house transaction data ([Hao and Michaud, 2024](#)).

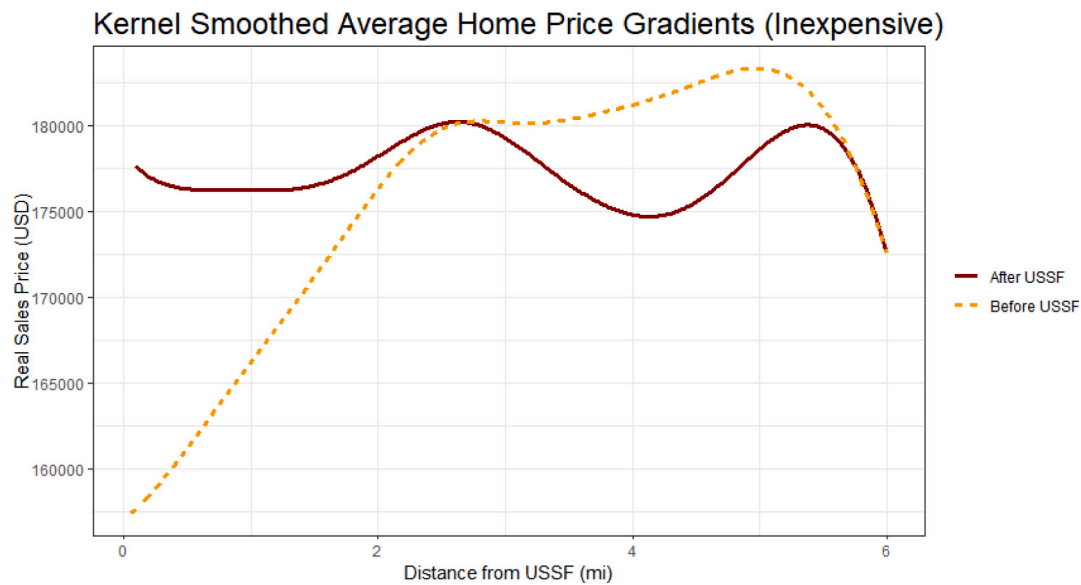


Fig. 4. This figure shows the price gradient with respect to distance from the USSF for the subsample of the less expensive (below the median) house transactions for both the pre-USSF and post-USSF eras. The pre-USSF and post-USSF curves do not align well at any point over the six mile radius studied here. This suggests that identification off of this subset of less expensive homes is less reliable than identification off of the more expensive subset of house transactions.

It is worth considering whether specific geographic factors may help explain why our results are somewhat larger than those reported in earlier studies. Our study area – located in the exurbs of the Richmond, Virginia metropolitan region – is relatively rural. Prior literature suggests that homes in rural communities may experience more pronounced declines in value following the construction of a utility-scale solar facility (USSF) than those in more urban settings. In this context, our comparatively larger estimates, relative to studies such as [Elmallah et al. \(2023\)](#) and [Dröes and Koster \(2021\)](#), could be interpreted as supporting this hypothesis. However, the similarity of our findings to those reported in the nationwide analysis by [Hu \(2025\)](#) complicates this interpretation. Further research that examines the house price effects of USSFs across the urban–rural spectrum would be valuable in assessing the robustness of this interpretation.

From a policy standpoint, one thing is clear. Constructing USSFs in isolated areas with few houses is better than constructing USSFs near more densely populated areas. However, this does imply constructing USSFs in rural areas where the literature suggests that negative house price effects are possibly slightly larger in magnitude. Thus, local governments may want to incentivize the construction of USSFs in the most isolated locations possible, possibly via location-contingent subsidies for developers. Moreover, for rural landholders who hold land that does not include built structures, the potential for USSF development on their properties will increase the value of their properties ([Abashidze and Taylor, 2023](#)). Importantly, as [Abashidze and Taylor \(2023\)](#) noted, this land value effect is not dependent on proximity to previously constructed USSFs, but rather is a consequence of proximity to the grid. This has further indirect local economic development benefits through increased tax revenues for local governments via higher assessment values, leading to increased revenues from property taxes. Increased tax revenues means an increased ability to improve or expand on existing infrastructure, which will perhaps improve the well-being of local residents in the future. Therefore, incentivizing USSF development in the most isolated parts of rural communities may be one way to maximize positive externalities while minimizing negative ones.

5.2. Housing turnover results

To understand the supply and demand mechanisms underpinning our results, we perform the first housing turnover analysis in the USSF

literature. We do this specifically to determine if there was a decrease in demand for houses, since a decrease in demand will put downward pressure on house prices. Thus, the final set of results presented in this paper pertains to housing turnover. We modeled housing turnover using Eq. (2) with the number of houses sold each year in a given census block as our measure of housing turnover. Results on housing turnover are presented in [Table 5](#), where the first two columns display results obtained using a standard difference-in-differences model, and the second two columns present results from our preferred distance-decay difference-in-differences model.

Here, we see consistent results across all models, indicating a decrease in housing turnover in census blocks closer to the solar facility. The baseline difference-in-differences model (columns 1 and 2) shows a statistically significant decrease of about 0.2 houses per year. Disaggregating this effect using our preferred distance-decay difference-in-differences model, we see that this effect was most prominent for census blocks between one and two miles from the facility. A decrease of approximately 0.6 houses per year is attributable to the construction of the USSF. The closest census blocks to the facility, those within one mile, experienced a decrease in turnover of about 0.3 houses per year, again highly statistically significant. While census blocks between two and three miles from the solar facility were the least affected by construction of the USSF, house turnover still decreased by about 0.1 houses per year, and this result was also statistically significant.

To ensure that out-migration was not a driver of this effect, we ran a difference-in-differences model to determine if the USSF caused people to leave the census tracts closest to the USSF. In this model, our dependent variable consisted of annual ACS estimates of census tract population, and we included the standard difference-in-differences interaction between treated and post indicators along with standard two-way fixed effects for census tract and year. We found no significant effect for the interaction term. Results from this analysis can be found in [Table 12](#) in the appendix.

Collectively, these results strongly support the hypothesis that prospective buyers view the solar facility as a disamenity, reflected by a decrease in demand. This decrease in demand drives down house prices, thereby serving as the mechanism through which the Caden Energix Rives Road Solar Facility decreased house prices in Prince George County, Virginia.

Table 5
Housing turnover regression results.

	Block-Level Annual House Sales			
	3 Miles	3 Miles	Decay Model	Decay Model
Treated $\times$ Post	-0.239*** (0.051)	-0.242*** (0.005)		
Treated ($0 \leq D \leq 1$) $\times$ Post			-0.249*** (0.078)	-0.310*** (0.037)
Treated ($1 \leq D \leq 2$) $\times$ Post			-0.498*** (0.105)	-0.572*** (0.114)
Treated ($2 \leq D \leq 3$) $\times$ Post			-0.075 (0.102)	-0.099** (0.047)
Tract $\times$ Year FE		✓		✓
Block FE		✓		✓
Observations	1,920	1,920	1,920	1,920
# Treated Units	696quad	696	696	696
Adjusted R^2	0.003	0.743	0.003	0.743

Results from the regressions where the dependent variable is the number of houses sold in each block-year. Cluster-robust standard errors are provided in parentheses with clustering occurring at the census tract level. Statistical significance is denoted by * for $p < 0.10$, ** for $p < 0.05$, and *** for $p < 0.01$.

6. Conclusion

Our results expand on the existing literature on the effect of USSFs by showing that suburban properties experience an attenuating effect, with effect sizes ranging from nearly 9% to 2.3% when a new USSF is built nearby. By studying housing turnover, we also show that decreased demand for houses near the solar facility is the driver of the decrease in house prices caused by the construction of a new USSF. We provide partial evidence that land values, separate from house values, increase after the construction of a new USSF. This is likely due to the expected increase in demand for land suitable for solar development now that the feasibility of solar development has been established and the decrease in land supply from constructing a new USSF.

Taken together, our results suggest that first, locating solar developments away from denser housing could mitigate some of the negative price effects the literature has observed, which this paper confirmed. Second, our results combined with the existing literature lead us to speculate that houses in more rural areas are the most likely to experience a net increase in value as an increase in land values due to the newfound feasibility of solar development. This can offset some – if not all – of the loss in value due to these disamenities. From the perspective of levying property taxes, targeting rural communities for solar development also seems ideal. Given that less valuable houses are more likely to be owned by less wealthy people, any policy incentivizing the location of solar development in rural areas comes with a tradeoff. An increase in the value of these properties is simultaneously a progressive wealth transfer via a regressive property tax adjustment. In the long run, owners of these houses may consequently become wealthier, assuming they can handle the increased financial burden of increased property taxes in the short run.

Some communities have initiated ordinances to control USSF development. Prince George County, Virginia, the housing market studied in this paper, established a limit on the proportion of land available for solar development to no more than 2.74% of the total land in the county. Perhaps due to the construction of the USSF studied in this paper, Prince George County is currently deliberating on restricting future USSFs to the county's M-3 Heavy Industrial Zoning district.⁵ Studies like ours can inform local economic policy makers on the local economic effects of USSFs. Our results are important because they allow for more nuanced local policy suggestions rather than broad mandates. Given our results, locating new solar developments in relatively more rural places appears to have several advantages relative to locating them in

relatively more urbanized areas. However, constructing USSFs in rural areas will come with tradeoffs. These facilities are often responsible for decreasing the remaining arable farmland (Hernandez et al., 2015). This potential loss of farmland may be unacceptable for communities that want to preserve their rural and agricultural heritage. Thus, no matter where future solar development occurs, the communities in which these facilities are built may need to be compensated. To address these issues, local policy makers may consider incentivizing solar developers to locate USSFs in isolated areas with subsidies or similar policy tools.

This study is not without its limitations. A case study of one specific housing market is not necessarily representative of the average effect of USSFs across all housing markets. However, case studies of specific housing markets, when aggregated, allow for a better understanding of how the distribution of house price effects from USSFs varies across the urban–rural continuum. Moreover, focusing on a specific case study allows minimizing the risk of bias from misspecifying an aggregated model of many housing markets which will inevitably not capitalize (dis)amenities in a homogeneous manner. And yet, in the case of this exurban community, our findings do indeed align with previous national studies, e.g., Hu et al. (2025).

Constructing a USSF has highly heterogeneous effects on property values over distance from the USSF. Specifically, we found a distance-decaying effect where the decrease in house prices was largest closest to the facility. Constructing a USSF may also have heterogeneous effects on property values across levels of both wealth and urbanization; however, the kernel density plot for the lower valued houses in our sample did cast some doubt on this possibility. Nevertheless, as solar development becomes more widespread as the industry seeks to play a major role in clean energy goals, local policymakers need to know the full spectrum of these effects in order to set optimal land use policies. This paper is a critical step in that endeavor. Future research can build on this paper by further exploring the urban–rural divide in price effects from renewable energy generation projects to more fully understand the economic effects of renewable energy generation and provide a more just transition to renewable energy.

CRedit authorship contribution statement

Michael Cary: Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Data curation. **Susan Chen:** Writing – review & editing, Supervision, Project administration. **Jessica Douthitt:** Writing – original draft, Investigation, Formal analysis, Data curation. **Njeri Jackson:** Writing – original draft, Investigation, Formal analysis, Data curation. **Michael McCloud:** Writing – original draft, Investigation, Formal analysis, Data curation. **Benjamin Rosner:** Writing – original draft, Investigation, Formal analysis, Data curation.

⁵ https://www.princegeorgecountyva.gov/planning_and_zoning/solar_energy_facilities.php

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Declaration of competing interest

The authors declare no conflicts of interest with this submission to Land Use Policy.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.landusepol.2025.107796>.

Data availability

All data and code are in an associated GitHub repo.

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